AWS CodeBuild / CodeDeploy / CodePipeline — Important Interview Questions & Answers

1.What is AWS CodeBuild?
AWS CodeBuild is a fully managed build service used to compile source code, run tests, and create deployment artifacts.
It automatically provides the compute environment required for the build process.
Simple ga gurthupettuko:
CodeBuild ante code ni build chesi tests run chesi deployment artifact create cheyyadaniki use chestham.
Eng: CodeBuild is a managed service used to build code, run tests, and create artifacts.

2.What is AWS CodeDeploy?
AWS CodeDeploy is a deployment service used to deploy application code to supported compute services such as EC2, Lambda, and ECS.
It helps automate application deployments and reduce manual deployment work.
Simple ga gurthupettuko:
CodeDeploy ante application code ni EC2, Lambda, ECS lanti services ki automatically deploy cheyyadaniki use chestham.
Eng: CodeDeploy automates application deployments to supported AWS services.

3.What is AWS CodePipeline?
AWS CodePipeline is a continuous delivery service used to automate the stages of an application release process.
It can connect source, build, test, and deployment stages into one pipeline.
Simple ga gurthupettuko:
CodePipeline ante Source → Build → Test → Deploy process ni automate chese service.
Eng: CodePipeline automates the application release process from source to deployment.

4.What is CI?
CI stands for Continuous Integration.
It means frequently integrating code changes into a shared repository and automatically building and testing those changes.
Simple ga gurthupettuko:
CI ante code changes frequently integrate chesi automatic ga build and test cheyyadam.
Eng: Continuous Integration automatically builds and tests frequently integrated code changes.

5.What is CD?
CD stands for Continuous Delivery or Continuous Deployment depending on the context.
It focuses on automatically preparing or deploying tested application changes to environments.
Simple ga gurthupettuko:
CD ante tested code ni deployment ki ready cheyyadam or automatically deploy cheyyadam.
Eng: Continuous Delivery or Deployment automates the process of releasing application changes.

6.What is the difference between CodeBuild and CodeDeploy?
CodeBuild is used to build and test application source code and create artifacts.
CodeDeploy is used to deploy the application to supported compute environments.
Simple ga gurthupettuko:
CodeBuild code ni build chestundi, CodeDeploy code ni deploy chestundi.
Eng: CodeBuild builds the application, while CodeDeploy deploys it.

7.What is the difference between CodeDeploy and CodePipeline?
CodeDeploy focuses on deploying an application to supported target environments.
CodePipeline manages the complete release workflow and can include source, build, test, and deployment stages.
Simple ga gurthupettuko:
CodeDeploy deployment kosam, CodePipeline complete CI/CD workflow kosam.
Eng: CodeDeploy handles deployment, while CodePipeline orchestrates the complete release workflow.

8.What is the difference between CodeBuild and CodePipeline?
CodeBuild performs the build and test operations.
CodePipeline coordinates different stages such as source, build, test, and deployment.
Simple ga gurthupettuko:
CodeBuild build chestundi, CodePipeline stages anni coordinate chestundi.
Eng: CodeBuild performs builds, while CodePipeline coordinates the release stages.

9.How does CodePipeline work?
CodePipeline starts with a source stage where application code is retrieved.
The code then moves through configured stages such as CodeBuild for building and CodeDeploy for deployment.
Simple ga gurthupettuko:
Source → CodeBuild → Test → CodeDeploy.
Eng: CodePipeline moves application changes through source, build, test, and deployment stages.

10.What is a Build Project in CodeBuild?
A Build Project defines how CodeBuild should run the build process.
It contains settings such as source, compute environment, build commands, environment variables, and output artifacts.
Simple ga gurthupettuko:
Build Project ante CodeBuild ela build cheyyalo define chese configuration.
Eng: A Build Project defines the configuration and commands used for a CodeBuild process.

11.What is buildspec.yml?
buildspec.yml is a configuration file that defines build commands and related settings for AWS CodeBuild.
It can specify phases such as install, pre_build, build, and post_build.
Simple ga gurthupettuko:
buildspec.yml ante CodeBuild lo build process and commands define chese file.
Eng: buildspec.yml defines the commands and phases for a CodeBuild project.

12.What are CodeDeploy Deployment Types?
CodeDeploy commonly supports In-Place and Blue/Green deployments for supported compute platforms.
In-Place updates existing instances, while Blue/Green uses a new environment before shifting traffic.
Simple ga gurthupettuko:
In-Place existing environment ni update chestundi, Blue/Green new environment prepare chesi traffic shift chestundi.
Eng: In-Place updates existing instances, while Blue/Green deploys to a new environment before traffic is shifted.

13.What is Blue/Green Deployment in CodeDeploy?
Blue/Green deployment creates or uses a new environment for the application version before shifting production traffic.
It helps reduce deployment risk and can support easier rollback.
Simple ga gurthupettuko:
Blue/Green ante new environment lo version deploy chesi tarvata production traffic shift cheyyadam.
Eng: Blue/Green deployment uses a new environment before shifting production traffic.

14.How would you create a simple CI/CD pipeline using AWS services?
I would configure a source repository, use CodeBuild to build and test the application, and use CodeDeploy or another deployment service to release it.
Then I would connect these stages using CodePipeline.
Simple ga gurthupettuko:
Source → CodeBuild → CodePipeline → CodeDeploy → Production.
Eng: I would connect source, build, test, and deployment stages using CodePipeline.

15.How would you troubleshoot a failed CodePipeline?
I would identify the failed stage and check its execution details and logs.
Then I would check the source configuration, build commands, permissions, artifacts, deployment configuration, and target environment.
Simple ga gurthupettuko:
Pipeline fail ayite Failed Stage → Logs → Source → Build → Permissions → Deploy check chestham.
Eng: I would check the failed stage, logs, permissions, artifacts, and deployment configuration.

16.What is the most important thing to remember about CodeBuild, CodeDeploy, and CodePipeline?
CodeBuild builds and tests the application, CodeDeploy deploys the application, and CodePipeline connects and automates the complete release process.
These services are commonly used together to build CI/CD pipelines on AWS.
Simple ga gurthupettuko:
CodeBuild = Build, CodeDeploy = Deploy, CodePipeline = Complete Pipeline.
Eng: CodeBuild builds, CodeDeploy deploys, and CodePipeline automates the release workflow.